

BANK MARKETING ANALYSIS REPORT

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Tools Used - Python, SQL Server, Power BI

INTRODUCTION

This report analyses a banking marketing campaign dataset to understand customer behaviour and identify factors influencing subscription decisions. The objective is to study patterns in customer demographics, campaign interactions, and economic indicators to improve future marketing strategies and increase subscription rates.

OBJECTIVES

- To analyse customer characteristics (age, job, marital status, etc.) and their impact on subscription decisions.
- To identify factors that influence whether a customer subscribes to a term deposit.
- To evaluate the effectiveness of marketing campaigns (number of contacts, duration, previous outcomes).
- To study how economic indicators (like employment rate, interest rate, etc.) affect customer responses.
- To build insights that help improve future marketing strategies and increase subscription success rate.
- To identify patterns of potential “yes” customers for better targeting.

BUSINESS PROBLEM

A bank is running telemarketing campaigns to encourage customers to subscribe to a term deposit product. The campaigns require significant resources, including call centre staff, time, and marketing budget. However, not all customers are equally likely to subscribe, and repeated calls can increase costs without improving results.

Key Business Question

"How can the bank maximize subscription conversions while minimizing marketing costs by targeting the right customers at the right time through the most effective communication strategy?"

KPI & Business Performance

1. What is the overall subscription rate of the campaign?
 2. Which month generated the highest number of subscriptions?
 3. Which contact method resulted in the highest conversion rate?
 4. What is the average number of campaign calls required per subscriber?
 5. Which customer segment (age, education, marital status) has the highest subscription rate?
 6. How does previous campaign outcome (pout come) affect subscription rate?
 7. What percentage of customers had housing or personal loans?
 8. Which day of the week generated the highest conversions?
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Statistical / Analytical Questions

9. How does call duration influence subscription probability?
10. Is there a relationship between age and subscription rate?
11. Does the number of campaign contacts affect conversion success?
12. Do previously contacted customers convert more than first-time contacts?
13. How do economic indicators (euribor3m, employment rate, etc.) affect subscription behaviour?

Business Insight

14. Which customer segments are the strongest predictors of subscription and why?
15. What customer profiles should be prioritized to maximize ROI in future campaigns?

DATASET OVERVIEW

- **Number of Rows** = 41,188
- **Number of Columns** = 21
- **Target Variable** = “y” or “Subscription”
- Dataset is **HIGHLY IMBALANCED** =
Subscribers = 4640 (11.26%), non-subscribers = 36548 (88.73%)
- **Missing values** = No missing values
- **Duplicate Rows** = 12 duplicate rows that contribute <0.1% of the data
- After **deleting Duplicate Rows** (Shape = 21* 41176)

- **Important Features** =

Campaign	Number of calls
pdays	Days since last contacted
y	Subscription status
Poutcome	Previous marketing campaign
Emp.var.rate	Employment variation rate
Cons.price.idx	Average change in prices
Cons.conf.idx	Confidence index
Euribor3m	Interest rate
Nr. employed	Number of employed people

DATA CLEANING

- Highly Imbalanced dataset

```
# Checking Imbalanced Dataset via %age
print(df['subscribed'].value_counts(normalize=True) * 100)
```

```
subscribed
no      88.734583
yes     11.265417
```

```
subscribed
no      36548
yes     4640
```

- No NULL values

- Duplicates

There are 12 duplicate rows in the dataset

```
# REMOVING DUPLICATE ROWS
df.drop_duplicates(inplace=True)
```

- Handling Unknown Values

```
# REPLACING UNKNOWN IN JOB COLUMN WITH MOODE
mode_value = df["job"].mode()[0]
df["job"] = df["job"].replace('unknown', mode_value)
```

There were unknown values in job, marital, education, housing loan, personal loan we replaced the unknown values with mode where the total unknown values are less than 5% of total dataset

- Handling Unknown Values - Poutcome

There are also unknown values in pout come but that contribute 86% of the data so if we apply mode to it then data analysis will not be accurate so we keep it as it is as another category

```
# Percentage of each poutcome category in the dataset
df["poutcome"].value_counts(normalize = True).round(2)

# we will not change non existent to mode as this contains 86% of the data we will treat it as a column.

poutcome
nonexistent    0.86
failure        0.10
success        0.03
Name: proportion, dtype: float64
```

- **Checking Outliers**

There are outliers in age, duration, campaign, pdays, previous. All the outliers can be possible except the pdays that person is contacted last 999 days ago which means almost 3 years so there are less chances that they will subscribe so we replace it with 0 or never contacted.

- **Creating a new column of Pdays where 999 is replaced with Never Contacted**

```
# REPLACING 999 WITH NAN IN PDAYS COLUMN
df['pdays'] = df['pdays'].replace(999, 'never_contacted')
```

0 can be misinterpreted as:

- contacted today
- 0 days since last contact

It creates wrong analytical meaning, so 0 is NOT safe for pdays interpretation.

- **Feature Engineering**

- **Age Group**

```
# CONVERTING AGE COLUMN INTO CATEGORICAL COLUMN
df["age_group"] = pd.cut(
    df["age"],
    bins = [0,19,39,59,100],
    labels = ["Teenager", "Adult", "Middle-aged", "Old"]
)
```

- **Duration Group**

```
# CREATING DURATION GROUPS
df['duration_group'] = pd.cut(
    df['duration'],
    bins=[0, 100, 200, 300, 500, 1000, df['duration'].max()],
    labels=[
        '0-100 sec',
        '101-200 sec',
        '201-300 sec',
        '301-500 sec',
        '501-1000 sec',
        '1000+ sec'
    ],
    include_lowest=True
)
```

- **Mapping Subscription Column**

```
# MAPPING THE SUBSCRIPTION COLUMN TO BINARY VALUES
df['y_new'] = df['y'].map({'yes': 1, 'no': 0})
```

Mapping Subscription column to numerical values so that we can easily get idea about the correlation of variables with Subscription.

SQL ANALYSIS

1. What is the overall subscription rate (y) of the campaign?

```
SELECT count(*) AS [Total Customers],  
       sum(y_new) AS [Total Subscribers],  
       (count(*) - sum(y_new)) AS [Non Subscribers],  
       ROUND(AVG(CAST (y_new AS FLOAT)) * 100, 2) AS [Subscription Rate]  
FROM [dbo].[Bank_data];
```

results Messages			
total customers	Total subscribers	non subscribers	subscription_rate
41176	4639	36537	11.27

INSIGHTS:

The dataset is HIGHLY IMBALANCED because the Subscription Rate is only 11.27%

2. Which month generated the highest number of subscriptions?

```
SELECT month,  
       sum(CAST (y AS INT)) AS [Subscribed]  
FROM [dbo].[Bank Data ]  
GROUP BY MONTH  
ORDER BY Subscribed DESC;
```

	month	Subscribed
1	may	886
2	aug	655
3	jul	648
4	jun	559
5	apr	539
6	nov	416
7	oct	315
8	mar	276
9	sep	256
10	dec	89

INSIGHTS:

In May, August and July people Subscribe more than the other months

3. Which contact method (contact) resulted in the highest conversion rate?

```
select contact, ROUND(AVG(CAST (y_new AS FLOAT)) * 100, 2) AS [Subscription Rate]
FROM [dbo].[Bank_data]
group by contact
```

	contact	Subscription Rate
1	cellular	14.74
2	telephone	5.23

INSIGHTS:

Customers who are contacted on cellular has more Subscription Rate

4. Which customer segment (age group, education, marital status) has the highest subscription rate?

```
SELECT ROUND(AVG(CAST (y_new AS FLOAT)) * 100, 2) AS [Subscription Rate],
age_group, education, marital,
count(*) AS [Total Customers],
SUM(y_new) AS subscribers
FROM [dbo].[Bank_data]
GROUP BY age_group, education, marital
ORDER BY [Subscription Rate] DESC;
```

	Subscription Rate	age_group	education	marital	Total Customers	subscribers
1	83.33	Teenager	basic.6y	single	6	5
2	50	Middle-aged	illiterate	divorced	2	1
3	50	Old	basic.6y	divorced	6	3
4	50	Teenager	university.degree	single	32	16
5	50	Old	illiterate	married	2	1
6	50	Old	professional.course	single	2	1
7	45.52	Old	basic.4y	married	290	132
8	44.74	Old	university.degree	divorced	38	17
9	43.55	Old	high.school	married	124	54
10	42.86	Teenager	basic.4y	single	7	3
11	42.86	Old	basic.4y	divorced	119	51
12	40	Old	basic.4y	single	10	4
13	39.29	Old	professional.course	divorced	28	11
14	38.17	Old	university.degree	married	317	121

INSIGHTS:

Teenagers who have basic education and are single has the highest Subscription Rate but the sample size is very small so this is not that much reliable the middle aged and old people are more reliable

High conversion BUT low volume → not always reliable

Adult/Middle-aged segments are more stable for targeting

5. What percentage of subscribers already had a housing loan or personal loan?

```
SELECT [housing_loan],  
       SUM(y_new) AS subscribers,  
       ROUND(SUM(y_new) * 100.0 / (SELECT SUM(y_new)  
                                   FROM dbo.Bank_data), 2) AS [%age of subscribers]  
FROM   dbo.Bank_data  
GROUP BY [housing_loan];
```

housing_loan	subscribers	%age of subscribers
0	2025	43.6500000000000
1	2614	56.3500000000000

```
SELECT personal_loan,  
       sum(y_new) AS [Total subscribers],  
       round(sum(y_new) * 100 / (SELECT sum(y_new)  
                                FROM   dbo.Bank_data), 2) AS [%age of subscribers]  
FROM   dbo.Bank_data  
GROUP BY personal_loan;
```

personal_loan	Total subscribers	%age of subscribers
0	3956	85
1	683	14

INSIGHTS:

56.35% of Subscribers have Housing Loan whereas Only 14% has Personal Loan

6. Which day of the week produced the highest conversion rate?

```
SELECT day_of_week,  
       ROUND(AVG(CAST (y_new AS FLOAT)) * 100, 2) AS [Subscription Rate]  
FROM   dbo.Bank_data  
GROUP BY day_of_week;
```

day_of_week	Subscription Rate
mon	9.95
fri	10.81
thu	12.11
tue	11.79
wed	11.67

INSIGHTS:

Thursday (12.11%) produces the highest Conversion

Python Exploratory Data Analysis (EDA) And Statistical Analysis

- Descriptive Statistical Summary of Dataset

	age	duration	campaign	previous	emp.var.rate	cons.price.idx	cons.conf.idx	euribor3m	nr.employed
count	41176.00000	41176.00000	41176.00000	41176.00000	41176.00000	41176.00000	41176.00000	41176.00000	41176.00000
mean	40.02380	258.315815	2.567879	0.173013	0.081922	93.575720	-40.502863	3.621293	5167.034870
std	10.42068	259.305321	2.770318	0.494964	1.570883	0.578839	4.627860	1.734437	72.251364
min	17.00000	0.000000	1.000000	0.000000	-3.400000	92.201000	-50.800000	0.634000	4963.600000
25%	32.00000	102.000000	1.000000	0.000000	-1.800000	93.075000	-42.700000	1.344000	5099.100000
50%	38.00000	180.000000	2.000000	0.000000	1.100000	93.749000	-41.800000	4.857000	5191.000000
75%	47.00000	319.000000	3.000000	0.000000	1.400000	93.994000	-36.400000	4.961000	5228.100000
max	98.00000	4918.000000	56.000000	7.000000	1.400000	94.767000	-26.900000	5.045000	5228.100000

INSIGHTS:

The average age is 40 years but max is 98 but it is possible that people with age of 98 years can be customer. Average duration is 258 but some potential customers make like to know more about the subscription so this is also not an outlier but shows potential subscribers who are interested. Campaign is also not an outlier because those people were called who are interested it is possible to get 56 calls.

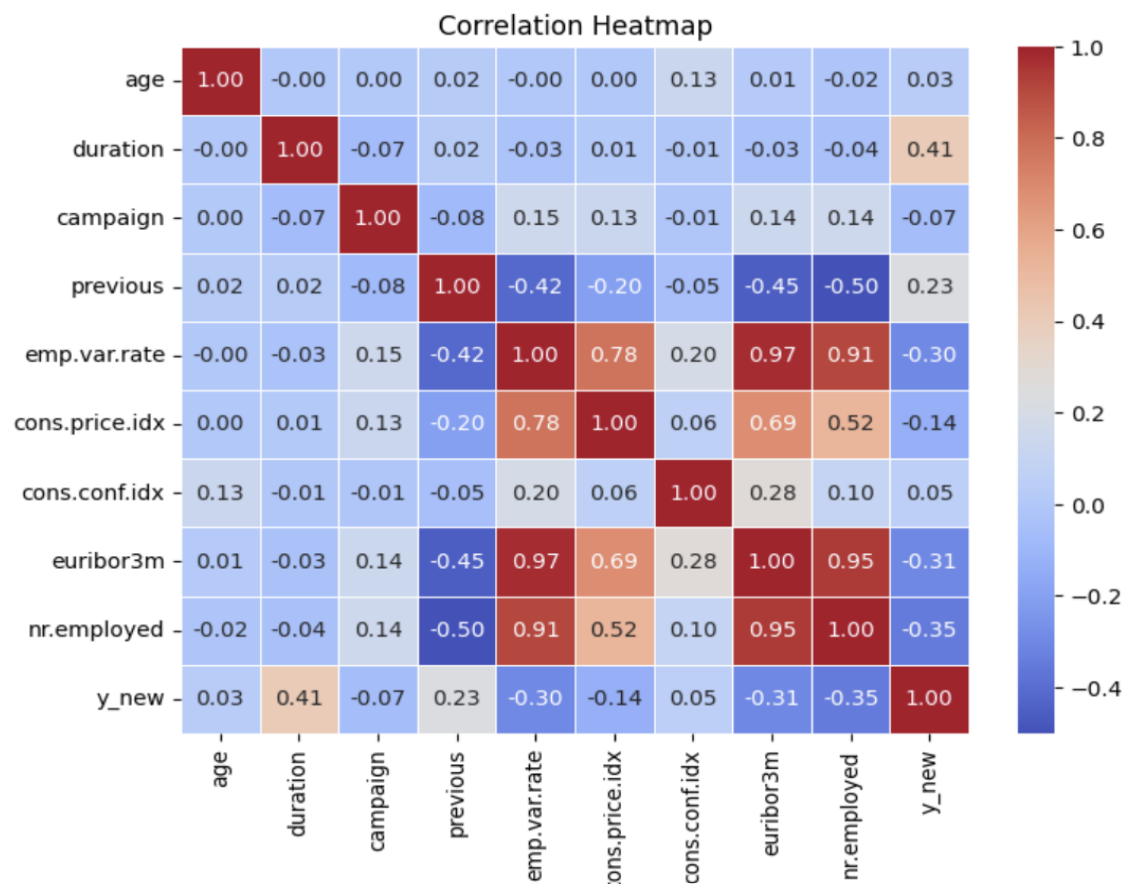
- Correlation Analysis

	age	duration	campaign	previous	emp.var.rate	cons.price.idx	cons.conf.idx	euribor3m	nr.employed	y_new
age	1.000000	-0.000808	0.004622	0.024379	-0.000242	0.001009	0.129075	0.010852	-0.017607	0.030381
duration	-0.000808	1.000000	-0.071765	0.020600	-0.027941	0.005303	-0.008126	-0.032861	-0.044672	0.405297
campaign	0.004622	-0.071765	1.000000	-0.079182	0.150786	0.127826	-0.013657	0.135169	0.144129	-0.066361
previous	0.024379	0.020600	-0.079182	1.000000	-0.420587	-0.203197	-0.050929	-0.454571	-0.501411	0.230202
emp.var.rate	-0.000242	-0.027941	0.150786	-0.420587	1.000000	0.775293	0.196257	0.972244	0.906949	-0.298289
cons.price.idx	0.001009	0.005303	0.127826	-0.203197	0.775293	1.000000	0.059170	0.688180	0.521945	-0.136134
cons.conf.idx	0.129075	-0.008126	-0.013657	-0.050929	0.196257	0.059170	1.000000	0.277864	0.100679	0.054802
euribor3m	0.010852	-0.032861	0.135169	-0.454571	0.972244	0.688180	0.277864	1.000000	0.945146	-0.307740
nr.employed	-0.017607	-0.044672	0.144129	-0.501411	0.906949	0.521945	0.100679	0.945146	1.000000	-0.354669
y_new	0.030381	0.405297	-0.066361	0.230202	-0.298289	-0.136134	0.054802	-0.307740	-0.354669	1.000000

INSIGHTS:

The correlation analysis shows that most variables have weak relationships with each other, indicating low multicollinearity in the dataset. **Call duration and Previous Customer Engagement** has the strongest positive relationship with Subscription, suggesting that longer customer interactions are associated with higher conversion rates.

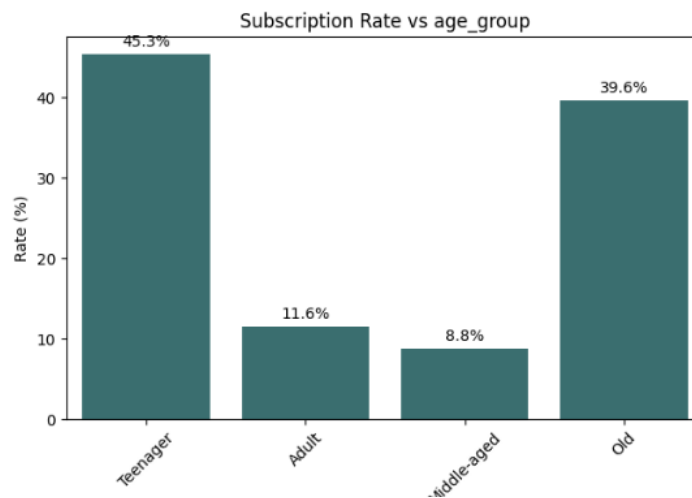
- **Correlation Heatmap**



- **INSIGHTS:**

Duration and **Previous** customer engagement has a **Positive** correlation with subscription whereas **Employment variation**, **Consumer Price Index**, **Euribo3m** and **nr.employed** has **Negative Relationship** with Subscription.

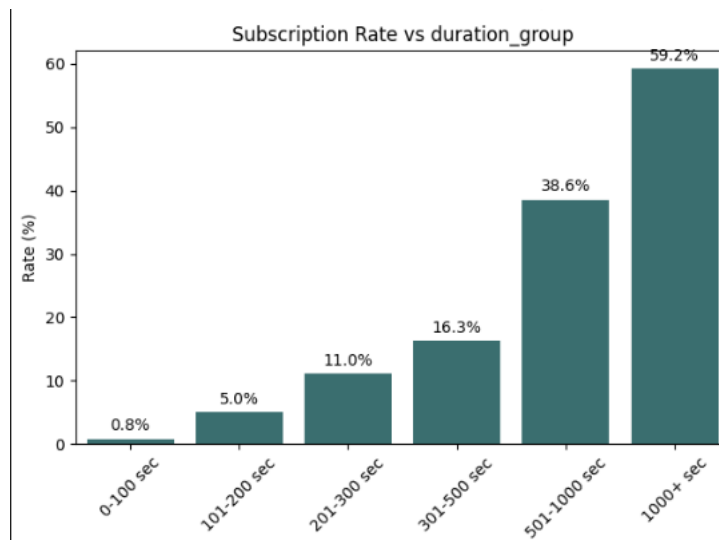
- **Age Group V/S Subscription Rate**



- **INSIGHTS:**

Maximum numbers of Subscribers are **Teenagers and Old** people but if we consider the sample size then the sample size for **Adults and Middle-Aged is more**, we have to consider this factor before coming to any conclusions.

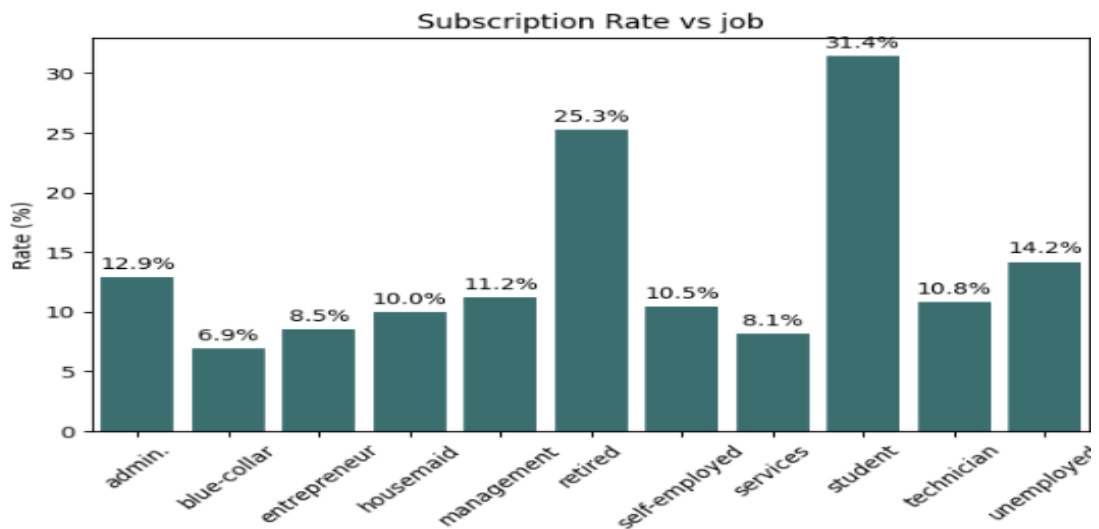
- **Duration V/S Subscription Rate**



- **INSIGHTS:**

The Longer the conversation the more chances of people to Subscribe.

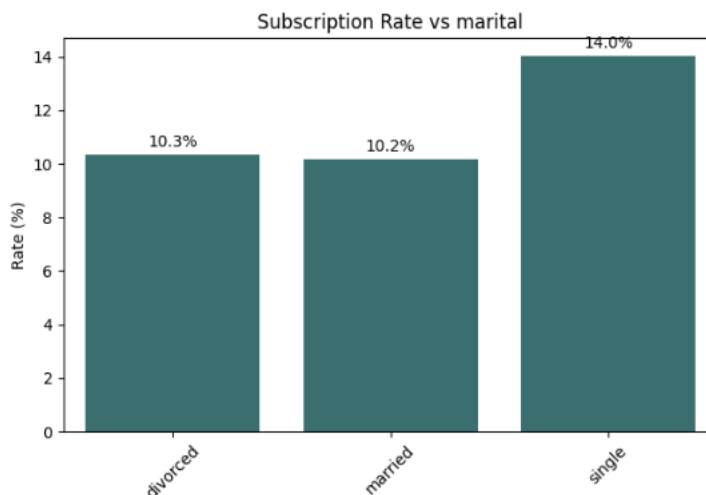
- **Job V/S Subscription Rate**



- **INSIGHTS:**

The maximum number of subscribers are students (31.4%) sample size (875) whereas the admin has maximum sample size (10749) and subscribers are (12.9%). So, we have to consider this thing also before concluding anything.

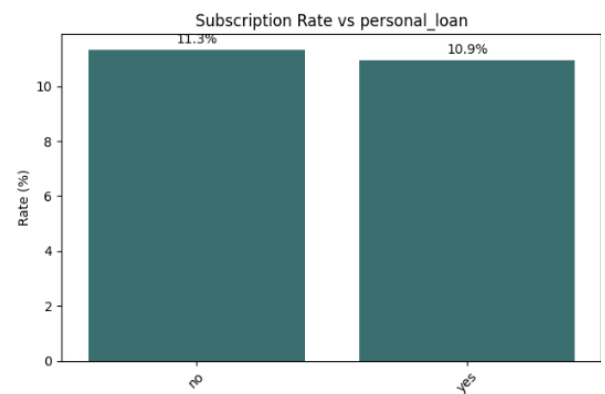
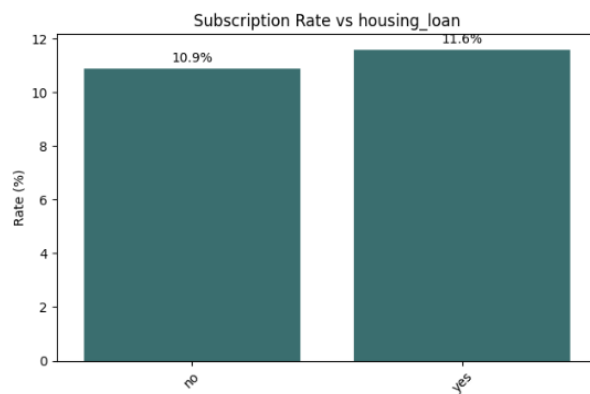
- **Marital Status V/S Subscription Rate**



- **INSIGHTS:**

The maximum number of subscribers are Single (14.0%) sample size (11,564) whereas the Married has maximum sample size (25,001) and subscribers are (10.2%). So, we have to consider this thing also before concluding anything.

- **Loan V/S Subscription Rate**

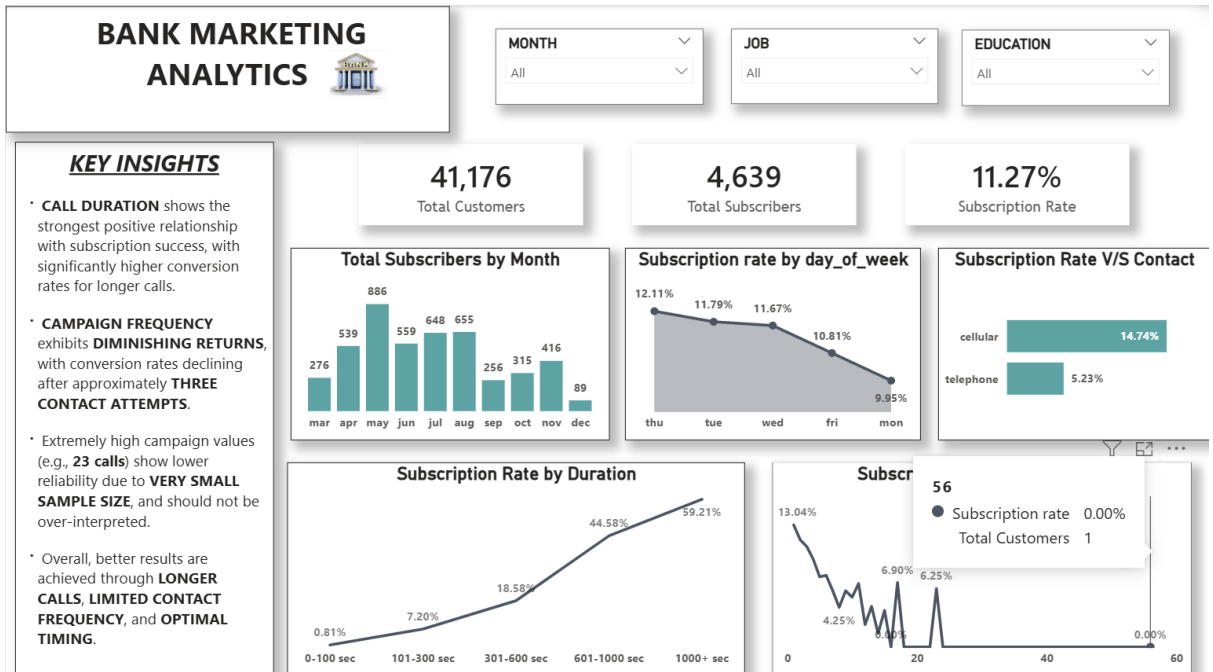


- **INSIGHTS:**

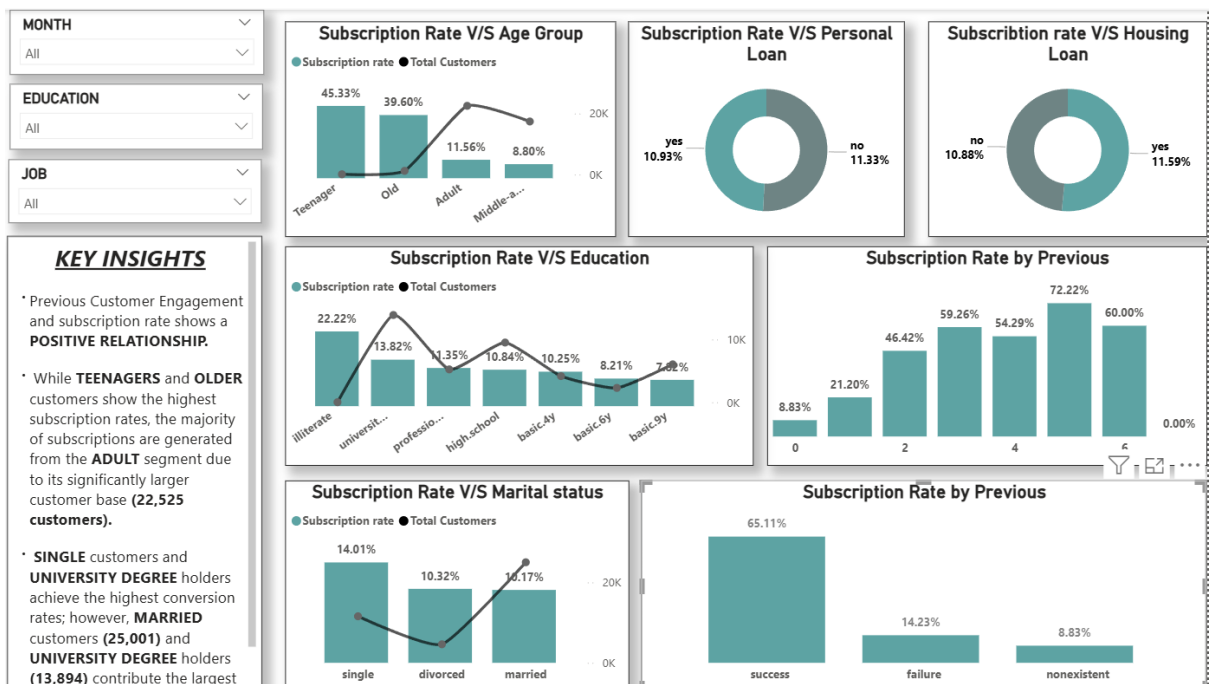
The maximum number of subscribers that have Housing Loan (11.6%) sample size (22,561) whereas maximum number of people who subscribed do not have Personal Loan (11.3%) and sample size (34,928).

Power BI Dashboard

Customer and Campaign Overview



Customer Segmentation

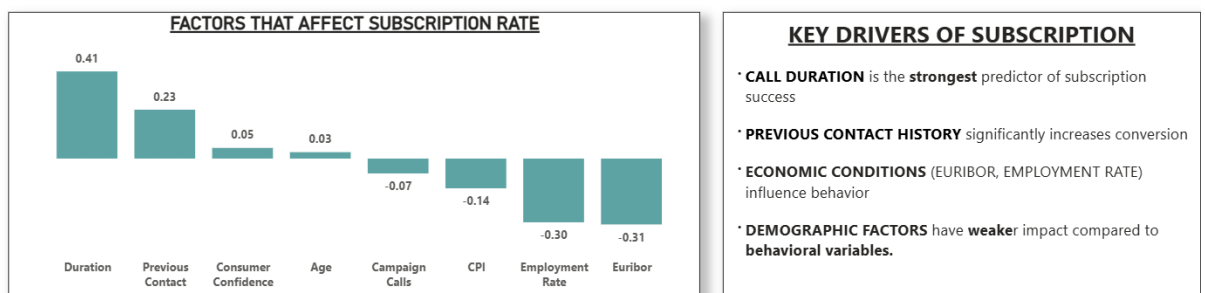


- ## Economic Factors



- ## Executive Summary

EXECUTIVE SUMMARY – BANK MARKETING ANALYSIS



BUSINESS CONCLUSION

- Subscription behavior is mainly driven by **BEHAVIORAL FACTORS**, especially **CALL DURATION** and **PREVIOUS CONTACT HISTORY**.
- **LONGER CALLS** and **REPEATED CONTACTS** significantly **INCREASE** conversion rates, showing strong customer engagement impact.
- **ECONOMIC CONDITIONS** (EURIBOR, EMPLOYMENT, CPI) **NEGATIVELY** affect subscription likelihood during stronger economic periods.
- **DEMOGRAPHIC FACTORS** have **LIMITED** influence, indicating targeting should focus more on engagement than customer profile.

Key Insights

The analysis shows that customer engagement, especially **call duration**, is the most important factor influencing subscription to term deposits. Customers with longer and more meaningful interactions are significantly more likely to subscribe.

Middle-aged and Adult customers with stable education levels form the most consistent and high-volume conversion segment, making them the most reliable target group for marketing campaigns.

Previous campaign outcomes and **Contact method** also play a key role, with successfully contacted customers and cellular communication showing higher conversion rates.

Overall, **Economic Indicators** has a **weaker impact** compared to direct customer interaction variables, highlighting the importance of targeted and efficient customer engagement strategies.

Business Recommendations

- Focus on customers with **longer call Duration**, as they show higher conversion rates.
- Target **Middle-Aged and Adult customers** with stable education levels for better campaign success.
- Prioritize **Cellular contact methods** over telephone for higher conversion rates.
- Re-engage customers with **successful Previous Campaign outcomes**, as they are more likely to subscribe again.
- Reduce unnecessary repeated contacts to improve **Campaign efficiency and ROI**.
- Optimize marketing efforts by focusing on **high-potential Customer Segments** instead of mass targeting.

Conclusion

- The analysis successfully identified key factors influencing customer subscription to term deposits.
- Customer engagement, especially **Call Duration**, emerged as the most important predictor of subscription success.
- **Middle-aged and Adult Customers** with stable education levels were found to be the most consistent and valuable target segment.
- **Cellular contact methods** and **positive previous campaign outcomes** significantly improve conversion rates.
- Economic indicators and campaign frequency had comparatively lower impact on subscription behaviour.
- The study provides actionable insights that can help the bank improve **Targeting Strategy, campaign efficiency, and ROI** in future marketing campaigns.

